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Application of Geographical Information System-Based Analytical Hierarchy Process Modeling for Flood Susceptibility Mapping of Krishna District in Andhra Pradesh

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Research Article

Keywords: Analytical Hierarchy Process (AHP), Digital elevation model, Flood susceptibility, Flood Hazard Index (FHI), Geographical Information System (GIS), Remote sensing, Sentinel-2A, Shuttle Radar Topography Mission

Posted Date: March 17th, 2022

DOI: https://doi.org/10.21203/rs.3.rs-1399020/v1

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Abstract

Flooding is one of the most catastrophic natural disasters in terms of provoking socioeconomic losses. The current study is to foster a flood susceptibility map of Krishna District in Andhra Pradesh (AP) through integrating remote sensing data, geographical information system (GIS), and the Analytical hierarchy process (AHP). Eleven factors including elevation, slope, aspect, land-use/land-cover (LULC), drainage density, topographic wetness index, stream power index, lithology, soil, precipitation, distance from the streams, are considered for identifying and evaluating the spatial distribution of critical flood susceptible regions. Thematic maps of different factors were derived in GIS using remote sensing data obtained from Sentinel-2A (satellite sensor), shuttle radar topography mission digital elevation model (SRTM DEM – v3), and other scientific data products. An analytical hierarchy process is a mathematical approach for decision support, primarily based on the weight and rank of different causative factors. AHP technique is implemented for flood hazard modeling and ascertaining the Flood Hazard Index (FHI) to produce a flood susceptibility map. Different thematic maps weighed with the AHP framework are combined using overlay analysis to produce the Flood Susceptibility map of the study region. The outcomes of the study demonstrate the potential of GIS and AHP in providing a premise to recognize the vulnerable areas that are susceptible to flood. According to the findings, the Flood Hazard Index is 42% and the study region is classified into high, moderate, and low susceptible respectively.

1. Introduction

Flooding often termed an inundation, is a Hydro-meteorological disaster that usually submerges dry land, is one of the most catastrophic natural hazards globally causing significant losses every year (Khosravi et al. 2019; Lawal et al. 2014; Mandal, Dharanirajan, and Sarkar 2021). It tends to happen, typically within one hour of precipitation in the form of heavy rainfall (Mohammadi et al. 2020; Zheng, Shanzhong, and Xu 2012; Petropoulos and Islam 2017), and results in human death, infrastructural destruction, environmental degradation, and loss of livelihood (Khosravi et al. 2019; P G 2013; Das 2020).

The flood events continue to increase every year as a result of several issues, including climate change and the rapid development of urbanization, civilization, and abnormal rainfalls are becoming much more prevalent today, posing tremendous flood risks and threatening infrastructure and livelihoods (Khosravi et al. 2019; Uddin et al. 2013; Yousefi et al. 2020; Shit et al. 2021).

Flooding-related infrastructure damage has led to a change in the water supply and inadequate service delivery in the fields of wastewater treatment, electricity, transportation, communication, education, and health care (Petropoulos and Islam 2017; Talha et al. 2019; Darwin et al. 2018; Hammami et al. 2019). Many flood protection acts have been launched and implemented but unfortunately, many developments have not been fully implemented (Patel and Yadav 2021; Natarajan et al. 2021). Therefore, it is necessary to predict the flood risk with the aid of imposing appropriate strategies within the study area to reduce the impact of the disaster.

Flood susceptibility of different regions across the world have been studied by different researchers (Ahmadlou et al. 2021; Khosravi et al. 2019; Patel and Yadav 2021; Petropoulos and Islam 2017; Talha et al. 2019), and several techniques and methods have been explored in order to simulate flooding (Asim, Nadeem, and Saima 2022; Daneshparvar et al. 2022; Seejata et al. 2018). Multi-Criteria Decision-Making analysis and machine learning techniques are adopted for flood susceptibility mapping to reduce losses arising from floods (Khosravi et al. 2019b; Swain, Singha, and Nayak 2020; Ramu et al. 2020). Some of the most powerful machine learning algorithms (Rahman et al. 2019) and auto encoder neural networks focused on reducing flood hazards include hybrid auto encoder-MLP (multilayer perceptron) (Ahmadlou et al. 2021a) and MLP neural networks. (Ahmadlou et al. 2021). Fuzzy mathematical models have been developed in order to enhance the probability estimate and evaluation of flood risk with inadequate data sets (Li et al. 2012). With the help of GIS and some analysis techniques, however, strategies such as Multi-Criteria Decision-Making analysis (Ibanga and Idehen 2020; Souissi et al. 2020; Allafta and Opp 2021; Dash and Sar 2020), machine learning, hybrid auto encoder-MLP (multilayer perceptron) (Ahmadlou et al. 2021a), MLP neural networks (Ahmadlou et al. 2021a), Frequency ratio method (Ullah and Zhang 2020), Fuzzy mathematical models (Ramu et al. 2020; Talha et al. 2019), and Artificial neural network techniques and processes produce the most precise findings and weights as close to reality as possible.

Because of the timing and location of occurrence, geophysical interactions, and geographical expanse of floods, it is not feasible to entirely manage floods (Patel and Yadav 2021; Demir and Kisi 2016). Complete safety and flood protection are rarely considered viable choices (Seejata et al. 2018; Pham et al. 2021; Hashim, Pour, and Misbari 2017). Advances in technology and remote sensing are essentially protecting us from terrible disasters and loss of life (Talha et al. 2019; Liuzzo, Sammartano, and Freni 2019; Bourenane, Bouhadad, and Guettouche 2019).

Remote sensing is the study of gathering data about an object or an event without making direct contact with it (Uddin et al. 2013; Asim, Nadeem, and Saima 2022; VILASAN and Kapse 2021). The Gis tool uses this data to do numerous analyses and expose the results to various needs via maps, reports, charts, and tables, which are provided in printed or digital format (Ibanga and Idehen 2020; Souissi et al. 2020; Ullah and Zhang 2020; Lawal et al. 2014; Aravinthasamy et al. 2021).

Flood estimating is a challenging and complex subject in hydrology (Elkhrachy 2015). It plays a vital role in reducing human tragedy and economic loss caused by floods (Nahiduzzaman, Aldosary, and Rahman 2015). The dependability of flood forecasting has improved in recent years as a result of advances in data collection via satellite observations, integration of hydrological and meteorological modeling capabilities for uncertainty analysis and communication (Razavi Termeh, Pourghasemi, and Alidadganfard 2018; Demir and Kisi 2016; Danumah et al. 2016). It can also serve as a crucial foundation for emergency services when it comes to implementing an innovative strategy to prevent and mitigate future floods.

Recently, the advancement of GIS tools and the availability of remote sensing data sources have substantially improved the formulation and implementation of prediction models for natural disaster-prone locations (Ravinder and Ramu 2020). Geospatial tools now enable a diverse set of data sources for high-quality flood simulation (Ravinder and Ramu 2020). Keeping this as an essential part the main objective of this study is to derive the extent of flood susceptibility zones in Krishna district, to evaluate which region needs the greatest involvement in the creation of risk reduction or mitigation techniques through the GIS based AHP modeling. The conclusions from this study may be utilized as a decision-making tool to improve flood hazard control plans and projects.

2. Study Area

Krishna District is located in the coastal area of Andhra Pradesh (16.6100° N, 80.7214° E). According to the 2011 Indian census, it has an area of around 8727 km2 (3370 sq mi), a total coastline of 88 km (55 mi), and a population of 4,529,009 people. It is surrounded on the east by West Godavari and on the south by the Bay of Bengal. The Krishna River, India's third longest, travels through the state of Andhra Pradesh before emptying into the Bay of Bengal at Hamsaladevi hamlet in the Krishna district. The district's climatic characteristics may be characterized as tropical, with extremely hot summers and fairly hot winters. The Southwest monsoons give roughly 1028 mm of annual rainfall to the region, and the area is covered with three types of soils: black cotton (57.6%), sand clay loams (22.3%), and red loams (19.4%). According to historical data from the study area, catastrophic flash floods hit various parts of the Krishna district in 1998, 2009, and 2019, causing substantial loss of life and property. As a result, flood susceptibility mapping is critical in this area for risk management and flood mitigation. The study area location map is shown in (Fig. 1).

3. Methodology

3.1 Data preparation

The methodology included in this study is depicted in the (Fig. 1). However, only a subset of factors was overlaid based on their significance in flood area selection. For inundation zoning, eleven thematic maps were considered, including elevation, slope, aspect, land-use/land-cover (LULC), drainage density, topographic wetness index, stream power index, lithology, soil, precipitation, distance from the streams, for identifying and evaluating the spatial distribution of critical flood susceptible regions.

In this study, the shuttle radar topography mission (SRTM) global 1 arc-second (30 m) DEM (digital elevation model) data is retrieved from USGS earth explorer using Krishna District coordinates with a cloud cover of less than 10%. The raw DEM data is imported into ArcGIS 10.3, and an elevation map of the research area is established. The slope % results were computed using the spatial analysis tool, and further, the DEM was then used to generate the aspect map and distance from the stream map for the area. Land-use/land-cover (LULC) was acquired using the map, which is created from ESA Sentinel-2 imagery at 10m resolution on a scale of 1/5700000 and mosaiced using Data management tools. The

research area is then extracted using the mask command. The USGS provides the lithology shape file for South Asia, and the lithology map for the research region is derived using the spatial analysis operation in Arc-GIS 10.3. From the FAO-UNESCO World Soil Map, the soil map is developed.

Preparing rainfall maps is essential for identifying flood-prone areas, but there are several steps to this process. The data was gathered from the APSDPS website, and then interpolated by selecting IDW. However, the resultant map of this research revealed that the biannual precipitation ranges between 751 and 1138 mm. These were categorized into four classes: 800 mm, 800–950 mm, 950–1100 mm, and > 1100 mm.

The Topographic Wetness Index (TWI) is a measure of wetness distribution that regulates water flow across the land. It's a standard measure that assesses the potential of water accumulating in certain areas. A high index number implies that there is a lot of water that may be collected because of the low slope, and vice versa. As a result, places with a higher TWI are more susceptible to flooding. The TWI of the study area was derived from the SRTM DEM. Finally, the Topographic Wetness Index (TWI) is computed using the Eq. 1,

$$TWI = \ln(\alpha / \tan\beta). \quad (1)$$

Where, α is the catchment area and β is the slope.

SPI, which is the product of the catchment area and slope, is used to calculate the erosive force of the overland flow. The SPI may be used to identify relevant locations for soil conservation measures, lowering the impact of concentrated surface runoff. Because of the increased stream power in the upstream stretch, the streams can erode and transfer a substantial amount of debris. When stream power is reduced, the channels become shallow and meandering, resulting in silt deposition on the overbank. On the lower plains, this is a major cause of floods. As a result, places with a lower SPI are more likely to flood. The SPI of the study area was derived from the SRTM DEM. Finally, the Steam power index is calculated by the Eq. 2,

$$SPI = \alpha * \tan\beta \quad (2)$$

Where, α is the catchment area and β is the slope.

Following that, thematic maps were created for each criterion by performing spatial analysis in GIS. These maps were reclassified using the weights derived from AHP technique and are utilized in overlay analysis to generate the Flood Hazard Index (FHI) for the entire district region.

3.2 Analytical Hierarchy Process (AHP)

The AHP approach is used to determine the impact of the flood on each element (Daneshparvar et al. 2022; Samanta, Pal, and Palsamanta 2018; Swain, Singha, and Nayak 2020), and a weight value equal to its relative relevance is applied to it. Saaty's pairwise comparison approach (Table 1) was used to calculate this weight, which he invented in 1980 (Das 2020; Hammami et al. 2019; Naik, Penki, and

SrinivasaRao 2019), was used to calculate this weight. An n*n dimensional pairwise comparison matrix is used to compare the conditioning factors. Each component in the matrix was assigned an arithmetic value between 1 and 9 based on its significance in relation to the other element. The arithmetic value 1 indicates that both components are equally important, however, value 9 indicates that the corresponding column factor is extremely important.

Table 1
Saaty's 1–9 scale for AHP (1980)

Conceptual Scale	Intensity of Importance	Inverse
Extremely importance	9	1/9
Very strong to extremely importance	8	1/8
Very strongly importance	7	1/7
Strongly to very strongly importance	6	1/6
Strongly importance	5	1/5
Moderately to strongly importance	4	1/4
Moderately importance	3	1/3
Equally to moderately importance	2	1/2
Equally importance	1	1

Table 2
Random index (RI)

n	1	2	3	4	5	6	7	8	9	10	11	12
RI	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49	1.51	1.54

A pairwise comparison matrix is created, and the consistency ratio (CR) is determined for the selected factors affecting the flood susceptibility using Eq. 3. If the CR is more than 0.1 (i.e., 10%), the judgments may be too inconsistent to be relied on. However, if the consistency ratio (CR) is smaller than 0.1, we may infer that our matrix is perfectly consistent and can proceed with the decision-making process utilizing AHP.

$CR = CI/RI$ – (3)

Where, CI is the Consistency index and RI is the Random index.

$CI = (\lambda_{max}\text{-}n)/n\text{-}1$ – (4)

n is number of factors and λ_{max} is average value of the consistency vector.

The eleven components were employed in this work for flood hazard mapping, and each parameter was separated into appropriate subclasses using reclassification and weights were assigned, as well as their normalized weight, were applied to estimate the flood hazard index using Eq. 5, the gathered data is analyzed in a GIS tool.

$$FHI = \sum_{i=1}^{n} w_i * r_i \quad -(5)$$

Where, n is number of parameters, w_i is the weighting factor, r_i is the rating of factors

$w_i = [(p_r * p_w)/v]*100$ – (6)

Where, p_r is the parameter rating, p_w is the parameter weight, and V is the summation value of the applied index

Using AHP, the flood susceptibility map was generated, to assess the relative relevance and get a ranking, the values in each row are compared to the values in each column. After filling up the pairwise comparison matrix (Table 3), the criteria are normalized to get the normalized matrix. Normalized values are calculated by dividing the value in each column by the total of the values in each column. Later criterion weights are derived by averaging all of the row's components. Following that, a consistency check is performed to determine whether or not the computed value is valid. The ratio is calculated by dividing weighted sum by criteria weights and the final flood susceptibility map was generated with a CR value of 0.02 (< 0.1 valid).

Table 3
Flood susceptibility via AHP method pair-wise comparison matrix

	1	2	3	4	5	6	7	8	9	10	11
1	1	1	3	4	6	7	5	2	8	7	0.5
2	1	1	2	3	4	4	3	2	8	7	0.33
3	0.33	0.5	1	2	3	4	2	1	5	4	0.2
4	0.25	0.33	0.5	1	1	1	1	0.33	3	3	0.2
5	0.17	0.25	0.33	1	1	1	1	0.25	3	2	0.14
6	0.14	0.25	0.25	1	1	1	0.5	0.25	2	1	0.17
7	0.2	0.33	0.5	1	1	2	1	0.33	4	3	0.17
8	0.5	0.5	1	3	4	4	3	1	5	5	0.33
9	0.12	0.12	0.2	0.33	0.33	0.5	0.25	0.2	1	1	0.12
10	0.14	0.14	0.25	0.33	0.5	1	0.33	0.2	1	1	0.11
11	2	3	5	5	7	6	6	3	8	9	1

Note: 1 = Slope, 2 = elevation, 3 = TWI, 4 = SPI, 5 = Drainage Density, 6 = LULC, 7 = Distance from the Streams, 8 = Aspect, 9 = Lithology, 10 = Soil, 11 = Precipitation

Finally, all of the parameters are converted to a raster format, with each raster layer's spatial resolution set to 30m x 30m cell size, and it is overlaid using a raster calculator in Arc GIS based on the AHP technique's effective weights. This provides an integrated database with three flood susceptibility classes: Low, Moderate, and High, which may be used to estimate the rate of flooding probability.

4. Results And Discussion

The flood susceptibility map (Figure) was generated using the effective flood susceptibility factors (Table 4) and the AHP Criteria Weights (%). The findings are addressed further below.

Slope map

The slope percent is a surface that is used to characterize the flood. A low-sloped region is more vulnerable since it is easily flooded by floodwaters. A surface with a steep slope, on the other hand, is less vulnerable to flooding since floodwaters may be swiftly drained down slope. As a result, this element is significant in determining the quantity of surface runoff and infiltration, and hence influences flood sensitivity. The research area is classified into five slope groups based on its slope (Table 4). The low

slope(0–30) indicates high Vulnerable and the high slope indicates(> 25^0) less high Vulnerable to flood (Fig. 3).

Elevation map

The elevation has a considerable influence on the propagation of floods. This parameter plays a significant function in controlling the following direction movement as well as the depth of the flood. The generated map was divided into five classes shown in the (Table 4). The low elevation(< 15) is high susceptible to floods and the high elevation(> 66) implies less susceptible to floods (Fig. 3).

TWI map

Because of the low slope, a high index value indicates that there is a lot of water that can be collected, and vice versa. As a result, areas with a higher TWI are more likely to get flooded (Fig. 3). In this study, TWI with (> 8.2) is assigned with a high risk, whereas TWI with (< 2.8) is associated with a lower risk (Table 4).

SPI map

The steam power index has negative values for locations with topographic potential for deposition and positive values for potentially erosive zones, with the maximum values associated with a steep slope gradient. This terrain form greatly contributes to the aggressiveness of erosion and the likelihood of land degradation. The lowest values (<-2.5) reflect generally flat terrain (Fig. 3), which influences high flood and sediment deposition and accumulation.

Drainage Density

The length of rivers per unit of area is expressed as drainage density, which is an important component of flood control techniques. The terrain of the study area has a high drainage density (> 15). On the other side, the plains have a low drainage density (0–2.5) (Table 4).

Land use and land cover (LULC) map

LULC is the most important element in determining whether places are at risk of floods. The rate of infiltration is influenced by land usage. Forests and plants, for example, promote water infiltration. Storm runoff is increased in urban areas, which are largely composed of impermeable surfaces and barren land. Water body, trees, grass, flooded vegetation, scrub, built-up area, and barren terrain were identified as LULC classes (Fig. 3).

Distance from the rivers map

The flood severity was maximum in the areas adjacent to the catchment outlet. As a result, the distance from the river is a crucial factor in flood mapping. 2000m, 2000-4000m, 4000-6000m, 6000-8000m, > 8000m are the distances from the river ranges, accordingly (Fig. 3).

Aspect map

Another contributing element in predicting flood occurrence is aspect, which is closely related to the convergence and direction of water flow toward which the slope of a mountain is facing (Fig. 3).

Lithology map

This component is connected to the permeability and storage capacity of the rock, which varies depending on the rock type. Rainfall water penetration is reduced when the rock is impermeable, enabling runoff surfaces to extend and enhance the flood event effect. The study area's Lithology map represents the undivided precambrian rocks(pc), quaternary sediments(q), cretaceous sedimentary rocks(ks), water(h2o) (Fig. 3).

Soil map

In the current study, thematic soil types were presented in a GIS layer assessing soils according to their textures and structures. Effectively, each soil type's texture and structure may have a significant influence on its permeability and, as a result, its water storage capacity. Water, pellic vertisols, dystric regosols, eutric nitosols, chromic livosols, thionic livosols, calcaric fluvisols, and lithosols are among the soil types illustrated on the map (Fig. 3).

Precipitation map

Heavy precipitation and extreme weather events have the most direct effect of increasing the intensity of other causes of flooding, resulting in more violent floods and greater flood risk. In the study area, biannual precipitation varies between 751 and 1138 mm. The higher precipitation (> 1100mm) leads to heavy floods.

Table 4
Flood susceptibility mapping parameters and sub-classes of different categories based on weights

Parameters	Sub-Class	Risk Vulnerability Rank	Parameter weight	Sub-Class weight (%)
Slope	0–3	1	0.185	49
	3–6	2		27
	6–13	3		14
	13–25	4		8
	> 25	5		2
Elevation	< 15	1	0.146	53
	15–32	2		25
	32–48	3		13
	48–66	4		6
	> 66	5		3
TWI	< 2.8	5	0.087	3
	2.8–4.2	4		5
	4.2–6	3		8
	6–8.2	2		17
	> 8.2	1		67
SPI	< -2.5	1	0.045	48
	-2.5 - -1.4	2		27
	-1.4 - -0.1	3		13
	-0.1–2	4		9
	2–12.6	5		3
Drainage Density	0–2.5	5	0.037	3
	2.5–5	4		7
	5–10	3		14
	10–15	2		24
	> 15	1		52
LULC	Water	1	0.031	30

Parameters	Sub-Class	Risk Vulnerability Rank	Parameter weight	Sub-Class weight (%)
	Trees	8		2
	Grass	7		4
	Flooded Vegetation	5		6
	Crops	3		15
	Shrub	6		5
	Built area	2		24
	Bare ground	4		14
Distance from the streams	< 2000	1	0.048	55
	2000–4000	2		27
	4000–6000	3		9
	6000–8000	4		6
	> 8000	5		3
Aspect	Flat	1	0.106	16
	North	6		9
	North east	2		14
	East	7		8
	South east	8		7
	South	5		10
	South west	4		12
	West	3		13
	North west	10		6
	North	9		5
Lithology	Undivided Precambrian rocks(PC)	4	0.018	9
	Quaternary sediments(Q)	3		16
	Cretaceous sedimentary rocks(Ks)	2		27

Parameters	Sub-Class	Risk Vulnerability Rank	Parameter weight	Sub-Class weight (%)
	Water(H2O)	1		48
Soil	Water	1	0.021	22
	Pellic Vertisols	3		15
	Dystric Regosols	6		10
	Eutric Nitosols	2		16
	Chromic Livosols	4		14
	Thionic Livosols	8		3
	Calcaric Fluvisols	7		7
	Lithosols	5		13
Precipitation	< 800	4	0.276	7
	800–950	3		17
	950–1100	2		35
	> 1100	1		41

The AHP approach was used to construct the final flood susceptibility map, which was divided into three broad groups with flood potentialities ranging from low to high (Table 5). According to flood susceptibility studies, high-class outlined regions have a high risk of flooding. It has a lower slope and elevation, a higher TWI, a lower SPI, is near to streams, has poorly drained soil and lithology, and receives extreme rain, all of which are critical conditions for flood mapping using AHP. According to this study, precipitation is the most important factor in floods. Since severe rain reduces the amount of infiltration. As a result, it flows into the river. The faster water travels into the river, the more likely it will flood. Water that travels over the surface (surface runoff) reaches a river faster than water that travels through the subsurface. So, the precipitation parameter was given the most weight (Table 4).

Table 5
Spatial distribution of flood susceptibility Classes

S.NO	Flood Susceptibility Class	Range	Area In Hectare
1	Low Susceptible	7.3–22.8	2,44,126.63
2	Moderately Susceptible	22.8–30.7	3,38,179.81
3	Highly Susceptible	30.7–42.1	2,43,241.34

In general, maps generated by integrated models addressed higher prediction accuracy and may be utilized for spatial prediction of flood hazard analysis in the research region. The results shown in (Fig. 4) of the model is categorized as low, moderate, or highly susceptible, indicating that the majority of flood-affected regions fall into the moderate and high categories. Table 5, shows the flood susceptibility map developed from the AHP technique which indicates that 243,241.34 ha of the study area is Highly Susceptible, 338,179.81 ha is moderately Susceptible and 244,126.63 ha of the area is less Susceptible to floods, for a total area of 825,547.79 ha, but the total area of Krishna district is 872,700 ha. This inaccuracy is triggered by the exclusion of the white shaded region shown in Fig. 4. Finally, the current findings of the study maps may be utilized as a reference for managing flood vulnerability in the Krishna district. Provincial governments can know about high-risk zones in order to adequately control flooding and arrange the progression of necessary flood prevention systems.

Flood Management plans

Particularly in the case of floods, Proposing different mitigation strategies play a vital role in reducing or eliminating the consequences of hazards. No infrastructure should be permitted in high flood plain regions, although structures with elevated platforms should be permitted in medium and low flood plain areas. In the case of a flood, residents in low-floodplain areas should have access to community infrastructure. Retrofitting river banks will be critical in reducing flood damage and increasing flood storage capacity. People are made aware of the need to take essential preparations before a flood occurs. Local governments and communities might employ a disaster management approach based on geospatial technology to make better decisions before and after the flood, minimizing the loss of life and property.

5. Conclusions

GIS and remote sensing methods and analysis are useful in a variety of disciplines. This has been utilized for mapping, modelling, and analysis of a vast scope of disaster management applications at various levels and stages. For flood vulnerability mapping in the Krishna district, a GIS-based spatial multi-criteria evaluation framework combined with the AHP approach has proven useful for demarcating flood hazard zoning. Elevation, slope, aspect, land-use/land-cover (LULC), drainage density, topographic wetness index, stream power index, lithology, soil, precipitation, and distance from streams were used as inputs to the AHP model for flood susceptibility mapping.

The use of remote sensing data in accordance with a GIS tool is extremely effective for preliminary flood hazard analysis. However, the availability of adequate datasets and data resolution is critical to the current method's reliability, performance, and applicability. The study employed an SRTM DEM with a spatial resolution of 30 m, which may be termed coarser for the floodplain region. Higher-resolution data could be used to fine-tune the study. The accuracy of the current technique might be improved by incorporating more detailed field data such as groundwater depth, local drainage, flood locations, and so

on. Furthermore, the lack of high-temporal-scale remote sensing data is a key impediment to flood mapping.

The AHP model's flood vulnerability zone was classified into three groups (high, moderate, and low). The findings clearly show that regions with low elevation and slope, high TWI, low SPI, high drainage density, and high precipitation are primarily risk-prone due to the high potential of flooding in such areas. The AHP technique used expert judgment and expertise to determine the best weights for the components that contribute to flood risk. The ease of handling, flexibility, and low cost of the applicable technique make it possible to employ, particularly in such a location with relatively limited data and information. In the present preliminary assessment of flood hazards, the AHP technique serves its purpose.

As, food risk management is a crucial segment of all social and environmental processes aimed at minimizing loss of life, injury, and property damage. The findings of this research will assist decision-makers in carrying out adequate flood management in future. Finally, the present study's maps may be utilized as a reference for flood prevention and preparedness by planners, developers, and the government. Knowledge of high-risk zones is essential for local governments to adequately control flooding and plan the execution of necessary flood prevention systems.

Declarations

Ethical approval: Not applicable

Consent to participate: Not applicable

Consent to publish: Not applicable

Author Contributions: All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by RAMU PENKI, SAI SANTOSH BASINA and SRINIVASA RAO TANNIRU. The first draft of the manuscript was written by SAI SANTOSH BASINA and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding: The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Competing Interests: The authors have no relevant financial or non-financial interests to disclose.

Availability of data and materials: Not applicable

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Figures

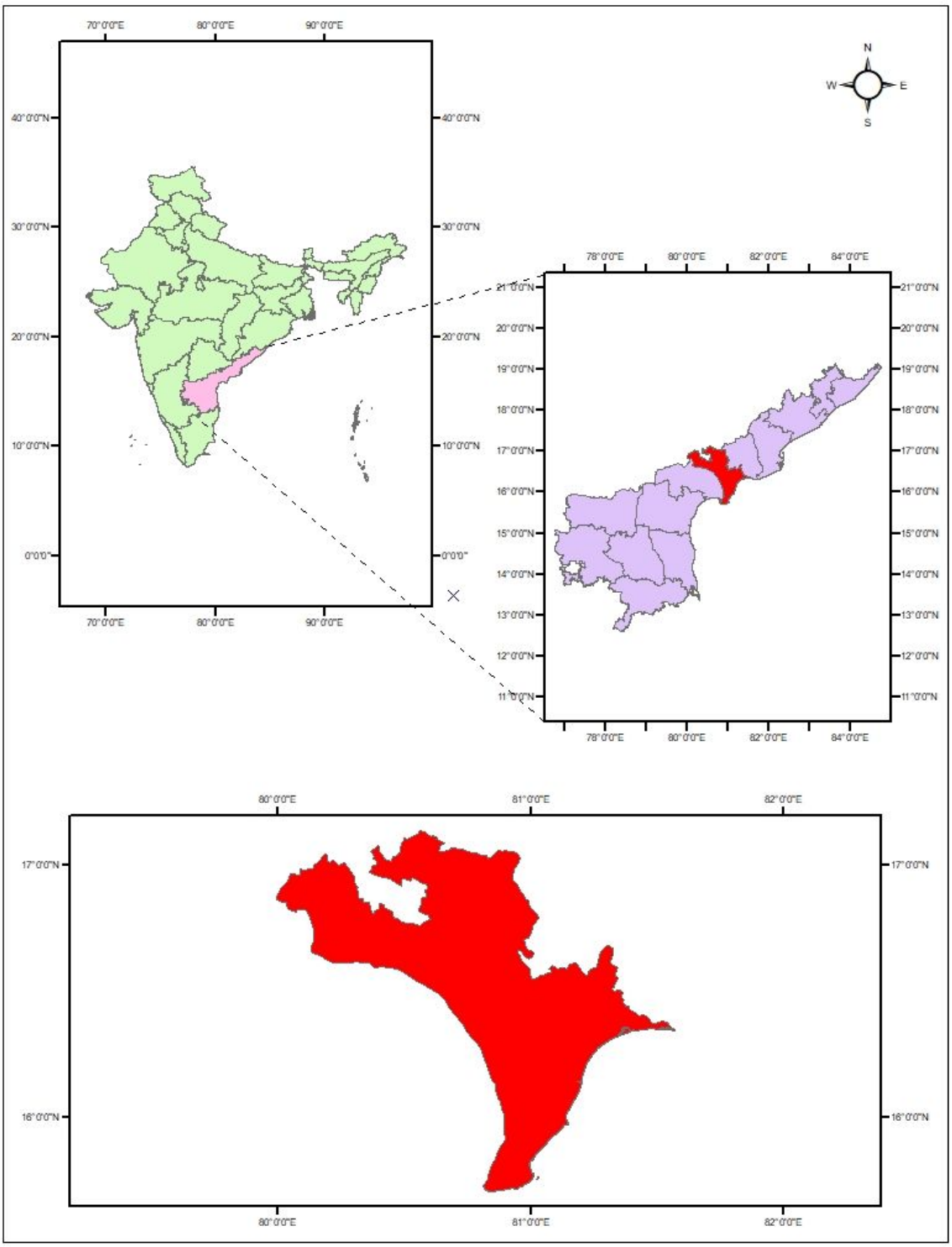


Figure 1

Study area location map

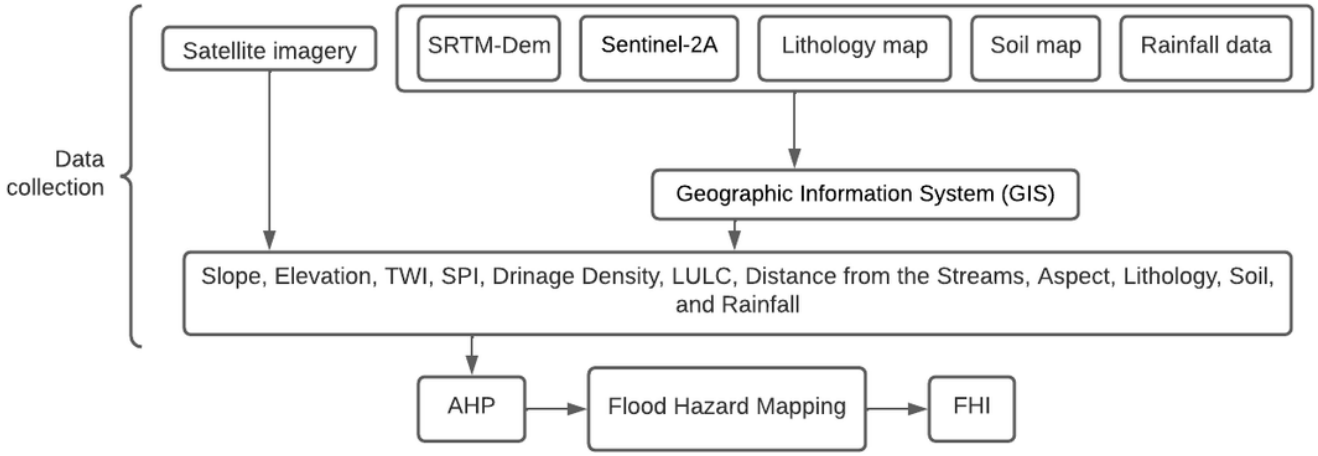


Figure 2

Arc-GIS conceptual model for flood susceptibility mapping based on the analytical hierarchy process.

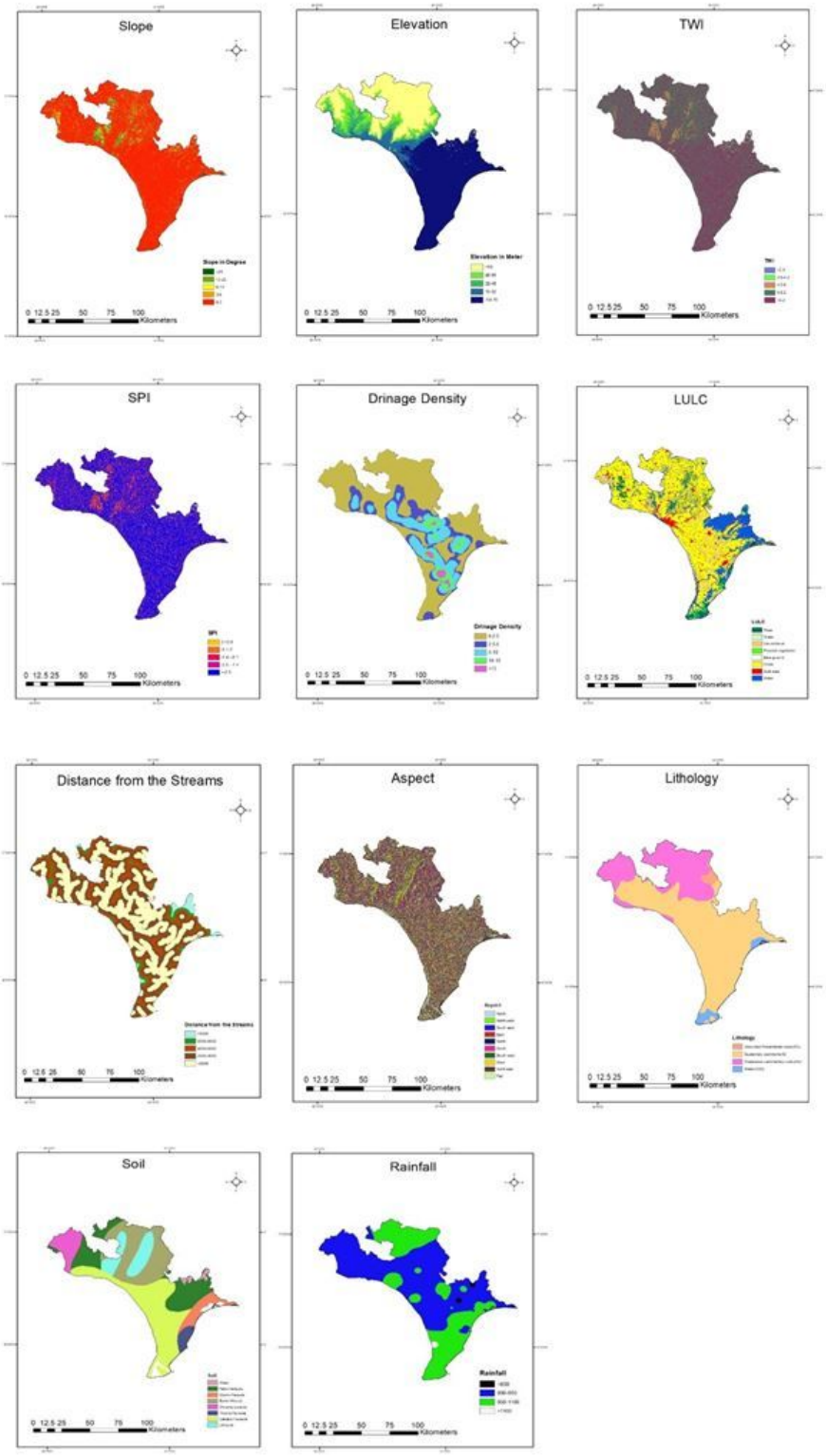


Figure 3

Parameters used for AHP modelling

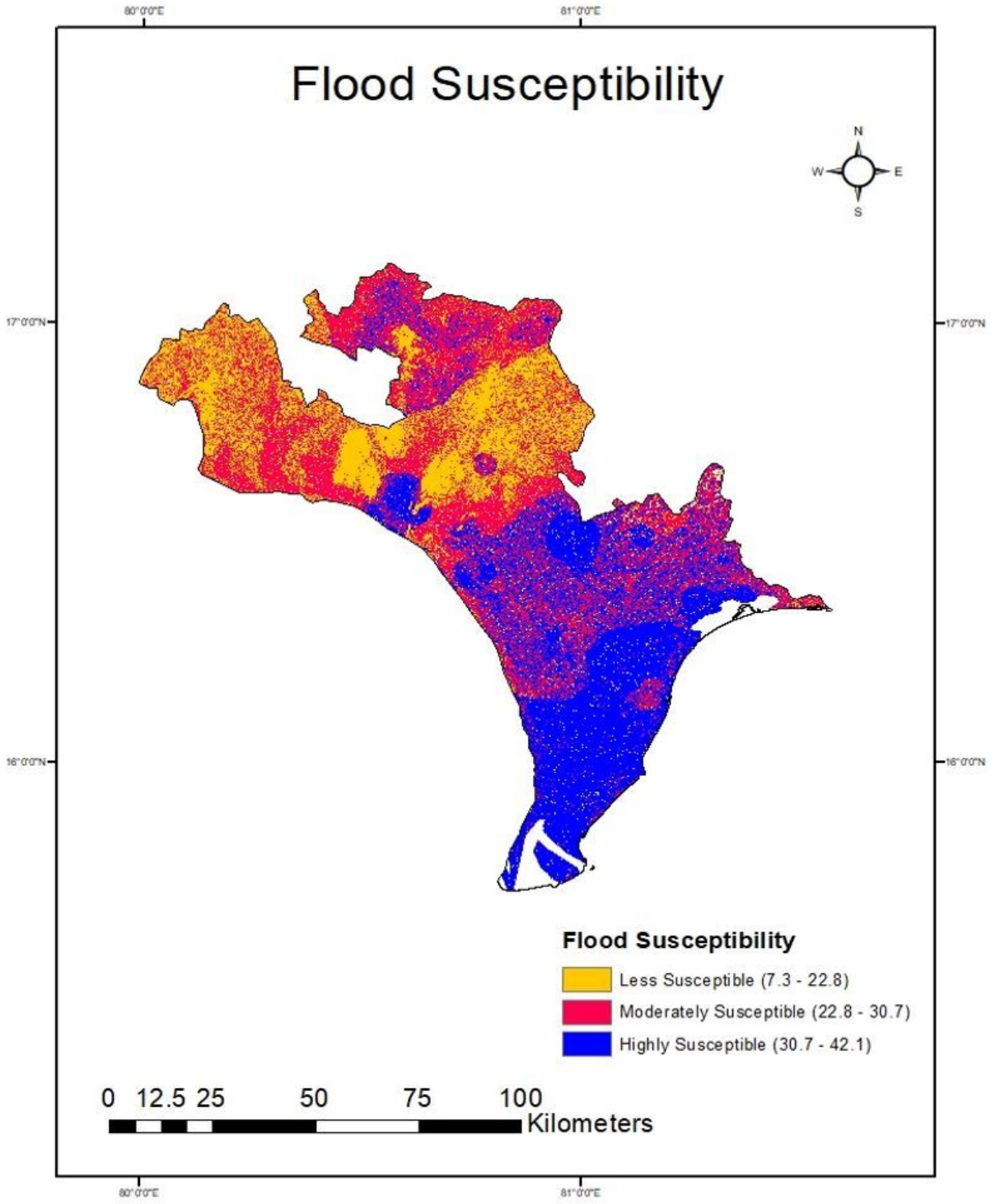


Figure 4

Flood susceptibility mapping of the study area